Sagnik Chakrabarti

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EDUCATION

Georgia State University *Atlanta, GA*

Master of Science in Computer Science

Bachelor of Science in Computer Science - Honors Magna Cum Laude

Relevant Coursework: Data Structures, Computer Organization and Programming, System Level Programming, Design and Analysis & Algorithm, Database Systems, Mobile App Development, Programming Language Concepts, Operating Systems, Web Programming, Advanced Computer Architecture, Artificial Intelligence, Computer Vision, Mobile Computing and Wireless Network Security and Machine Learning.

TECHNICAL SKILLS

Java, Python, Scala, SQL, C++, HTML, CSS, JavaScript, React(.Js/Native), Dart, Azure, Logic, Flutter, Terraform, Docker, Firebase, Google Cloud Platform, PHP, Spring-Boot, Linux, Angular, Svelte Kit, SwiftUI, Redis, Ansible, Splunk, Cassandra, Git-Ops, DynamoDB, Scripting, Django, Jira, AWS.

EXPERIENCE

Graduate Research Assistant **August 2025 – Present**

National Science Foundation (NSF-CoSEA Teacher Portal), Computer Science Department at Georgia State University *Atlanta, GA*

- Architected **Flask/PostgreSQL** platform serving **2K+ institutions** processing **100K+ daily submissions**; decomposed monolithic schema into **3NF model** reducing **latency p50 40%** (420ms to 250ms) and **p95 88%** (1.2s to 140ms) via **composite B-tree indexes**.
- Engineered **PostGIS R-tree** spatial engine with **Douglas-Peucker simplification** achieving **sub-100ms GeoJSON** rendering, deployed **Gunicorn (4 workers)** behind **Nginx** with **ETL pipeline** reducing full-table scans **45% to 7%** through strategic indexing.

Graduate Teaching Assistant (formerly, Undergraduate Teaching Assistant) **September 2024 – August 2025**

Computer Science Department at Georgia State University *Atlanta, GA*

- Led instruction of **90+ students** across 3 semesters in **SOLID principles**, polymorphic type systems, **AVL/2-3-4 trees**, graph algorithms (**Dijkstra, Bellman-Ford**), amortized analysis via aggregate method and accounting technique.
- Engineered automated grading using **Python AST** introspection with **pytest** parametrized fixtures and hypothesis **property-based testing**, instrumented **cProfile/memory_profiler** yielding runtime breakdowns for **500+ submissions**.

Computing and Informatics Center Tutor **September 2023 – May 2025**

Research Innovation and Scholarly Excellence (RISE) Grant Department at Georgia State University *Atlanta, GA*

- Conducted **200+ sessions** teaching top-down memoization, bottom-up tabulation with space-optimized rolling arrays, divide-and-conquer via **Master Theorem**, greedy exchange arguments with correctness proof techniques.
- Facilitated code reviews emphasizing correctness proofs and complexity lower bounds, median **30% reduction** in off-by-one errors, **75% successful refactoring** from **$\Theta(n^2)$ to $\Theta(n \log n)$** merge-based or $\Theta(n)$ hash-based solutions.

Inventory Analytics Team Associate Internship **April 2023 – August 2023**

University Housing at Georgia State University *Atlanta, GA*

- Built OLAP analytics pipeline processing **50K+ maintenance records** using **PostgreSQL window functions**, **recursive CTEs**, and **materialized views** with incremental refresh strategy across **15 residential facilities**.
- Executed zero-downtime migration of **5GB database** from **MySQL 5.7 to AWS RDS PostgreSQL 13.x** using parallel restore **(pg_dump --format=custom, jobs=4)**, optimized queries via prepared statement caching **reducing latency 60%**.

PROJECTS

Market Sphere - AI-Powered Paper Trading Platform | *React, FastAPI, Azure, Databricks, PostgreSQL*

- Designed microservices on **Azure Kubernetes (HPA: CPU=70%, min=2, max=10)** with **React 18 concurrent rendering**, **FastAPI async** backend via **uvloop (2x throughput)**, **WebSocket** handling **1K+ messages/sec** for live market data streaming.
- Implemented **3NF schema** with **Databricks Spark (8 nodes)** analyzing **1M+ OHLCV records** via **Z-ordered Delta Lake**, **JWT RS256 authentication** with token rotation, CSP/CORS policies achieving **zero security violations** in production.

MCP-Powered Data Lake Agent | *Python, Azure OpenAI, GPT-4, PostgreSQL, FAISS*

- Architected **RAG pipeline** using **SentenceTransformer (384-dim)** with **FAISS IVF quantization** achieving **89% recall@5** vs **68% TF-IDF baseline**, **GPT-4** few-shot prompting with schema injection, **ANTLR4 parser** ensuring SQL correctness.
- Implemented **row-level security** via WHERE clause injection preventing **PII leakage**, **Redis cluster (3M/3R)** with **SimHash LSH** semantic caching achieving **60% hit rate** reducing GPT-4 latency **800ms->120ms** through query deduplication.

SHADOW: Preventing RowHammer in DRAM with Intra-Subarray Row Shuffling| *C++, Verilog, DDR5, Ramulator*

- Architected **RNG-based intra-subarray row remapping** via **RFM interface** achieving **10^6× bit-flip reduction**, **<1% area overhead**.
- Simulated **DDR5 microarchitecture** under **SPEC CPU** + **blast attacks** validating **<3% throughput impact** vs **TRR/PARA baselines**.

EchoVibe - Mood-Based Music Recommendation App | *Flutter, Dart, FASTAPI, Python, Transformers, Firebase, Firestone*

- Fine-tuned **DistilBERT (6-layer, 66M params)** on **GoEmotions (58K samples, 28 labels)** achieving **0.74 macro-F1** via **fp16 training**, mapped emotions to Spotify features using **Russell's circumplex model** with **cosine similarity** over **12-dim vectors**.
- Deployed **FastAPI on uvicorn** (4 workers), Firestore with **composite indexes** (userId, timestamp DESC) optimized for read-heavy workloads, FutureBuilder with StreamController debouncing (300ms) maintaining **60 FPS UI rendering**.